ENGINEERING CONCEPT · ENGLISH TRANSLATION

How to Build a Single-Channel ECG Device

A technical outline based on the supplied requirements, from bench prototyping to medical-device verification.

ADS1291 | ADS1292ECG-FE | ESP32-S3

Scope: Preliminary engineering proposal translated from the original Russian text. Bench tests use simulated signals; the design is not a verified medical device.

Editor's note: Formatting of code and diagrams has been restored from the copied source. Nontechnical chat interface thumbnails have been omitted.

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Thirteen sections covering hardware, software, testing, and regulatory considerations.

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> This document is the original engineering concept, distinct from the later 32-card dialectical educational outline. Proposed limits and example parameters are not the same as verified medical-device specifications.

Original technical concept — English translation

The provided technical requirements call for the development of a portable, single-channel electrocardiograph that records an ECG, calculates heart rate, displays the waveform in real time, saves each examination, and transmits it to a physician or a medical information system.

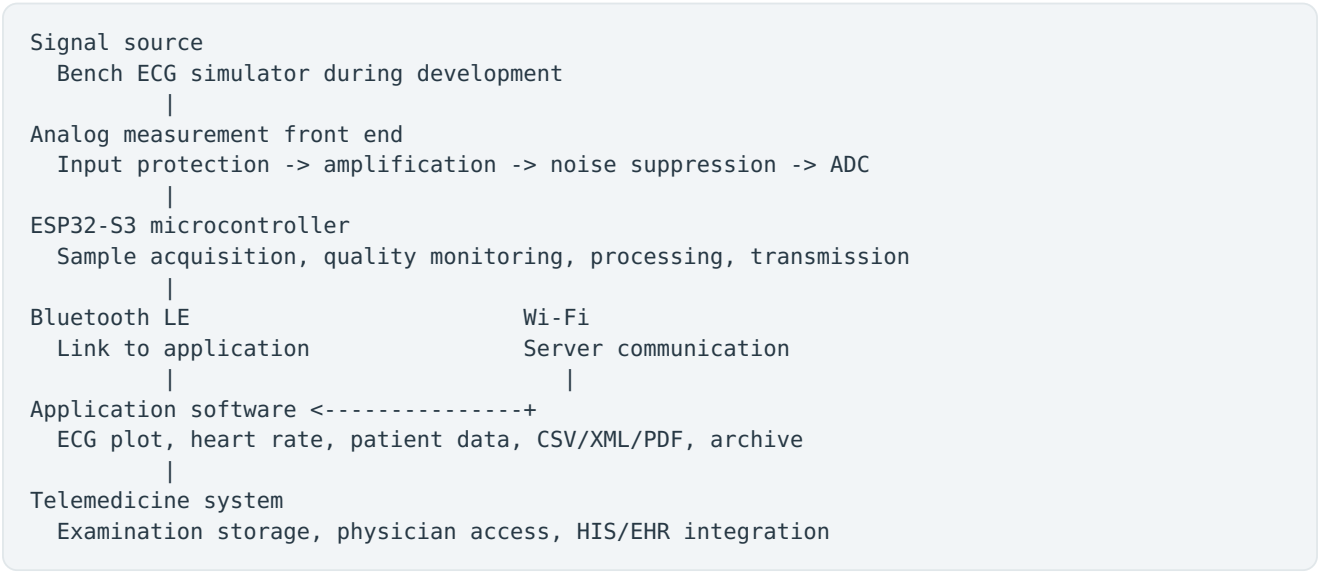
The requirements specify an analog-to-digital converter (ADC) with a resolution of at least 16 bits, a proposed sampling rate of at least 500 Hz, recordings lasting 10–30 seconds, noise filtering, electrode lead-off detection, battery power, and data transmission.

Important: These are preliminary functional requirements, not a complete electrical specification. They do not define numerical values for the input range, input impedance, common-mode rejection ratio (CMRR), allowable noise, battery life, or measurement accuracy. Those values must be agreed upon before a final PCB is developed.

I propose implementing the device in two stages: first, build a fully functional bench prototype using an ECG simulator; then, develop a protected measurement front end and complete the necessary electrical safety verification before any use on a person. An ordinary development board, even one with a battery and Bluetooth, is not in itself a safe medical electrocardiograph.

1. Overall Device Architecture

The system consists of the following functional stages:



The basic principle is that the measurement electronics convert a very weak electrical signal into digital samples; the microcontroller delivers those samples reliably; and the application displays and stores the data together with the acquisition parameters.

2. Selecting the Electronic Components

2.1. Main measurement IC — ADS1291

I would use the **Texas Instruments ADS1291** as the foundation.

The ADS1291 is a specialized, single-channel, 24-bit ADC for biopotential acquisition. It includes an amplifier with programmable gain, lead-off detection functions, and internal test signals. It supports conversion rates from 125 to 8,000 samples per second and the SPI interface.

Why this device:

- Its single physical measurement channel matches the requirements.
- It offers 24-bit ADC resolution rather than the required 16 bits.
- Its sampling rate can be set to 500 Hz.
- There is no need to assemble a separate low-signal amplifier from several operational amplifiers.
- It provides tools for self-testing the measurement front end.

However, a 24-bit ADC output does **not** mean 24 bits of real-world measurement accuracy. The effective resolution is limited by noise, the amplifier, electrodes, and acquisition conditions.

2.2. Board for initial development

For the bench stage, it is more convenient to use a ready-made **ADS1292ECG-FE** evaluation board.

It uses the two-channel ADS1292, but one channel can be disabled. The board already includes an MSP430F5529 microcontroller, USB, and the measurement front end. Texas Instruments supplies software for signal analysis, display, and export of test data.

Official product page: TI ADS1292ECG-FE.

An important technical detail: the board's original software was designed for Windows XP/7, so compatibility with Windows 11 is not guaranteed. Texas Instruments also explicitly limits the board to laboratory prototyping, not medical use.

2.3. Microcontroller

An **ESP32-S3** can be used for the custom portable version.

It performs the following functions:

1. Controls the ADC over SPI.
2. Receives digital signal samples.
3. Buffers recordings in memory.
4. Monitors sample loss and measurement status.
5. Transmits data over Bluetooth Low Energy or Wi-Fi.
6. Controls indicators and works with dedicated battery charging controllers.

The microcontroller should **not** measure the weak ECG using its own built-in ADC. In the proposed architecture, all precision digitization is performed by the ADS1291.

2.4. Complete list of functional units

Component	Quantity	Purpose
ADS1291	1	Analog measurement front end and 24-bit ADC
ESP32-S3	1	Control, buffering, and wireless transmission
Analog supply regulator	1	Low-noise power for the measurement front end
Digital supply regulator	1	Power for the microcontroller and interface
Protected battery module	1	Autonomous power
Battery charging controller	1	Charging and power-state management
USB-C	1	Charging and service interface
Control buttons	1–2	Start/stop recording
Status LEDs	2–3	Power, recording, fault indication
Nonvolatile memory	As calculated	Store records when no link is available
Bench-test connector	1	Connect the ECG simulator
PCB and enclosure	1 set	Mount and protect electronics
Medical-grade input protection	1	To be engineered separately for the future device
Certified electrodes and cables	1 set	Only for a version that has passed safety assessment

The laboratory prototype also requires a computer, a USB cable, a multimeter, possibly an oscilloscope, and a specialized ECG simulator to test the external measurement path.

3. Connecting the Electronic Blocks

3.1. Digital section

The ADS1291 and ESP32-S3 exchange data through SPI.

ADC signal	Function	ESP32-S3 side
SCLK	SPI clock	SPI CLK
DIN	ADC commands and configuration	SPI MOSI
DOUT	Measurement output	SPI MISO
CS	Chip selection	GPIO configured as chip select

ADC signal	Function	ESP32-S3 side
DRDY	New sample ready	GPIO with interrupt
RESET	Reset of measurement front end	GPIO
START	Start/stop conversion	GPIO

This table shows the **logical interfaces**, not a wiring diagram for a specific board. Actual pins, voltage levels, and start-up sequence must be taken from the selected board's documentation. The ESP32-S3 digital inputs must not be connected directly to 5 V USB power.

The most important signal is `DRDY`: it tells the controller that another conversion result is ready. The implementation must not rely solely on a software delay of exactly 2 ms, because scheduler, Bluetooth, and Wi-Fi delays can cause missed measurements.

The intended flow is:

```
ADC completes a conversion
          |
         DRDY
          |
Microcontroller interrupt
          |
Read result over SPI
          |
Assign sample index
          |
Write to buffer
          |
Separate data-transfer task
```

Buffering and wireless transmission must be decoupled. If Bluetooth temporarily delays a packet, the next ADC sample must still be acquired.

3.2. Analog section

The analog front end follows this functional chain:

```
Measurement inputs
          |
Electrical protection
          |
RF-interference protection
          |
Differential AFE input
          |
Programmable gain
          |
ADS1291 ADC
          |
Digital samples
```

For medical use, the external electrode interface and the other circuits must have patient-protection measures appropriate to the device category, including leakage-current control and the necessary isolation.

Designing this section is not simply a matter of connecting electrodes to IC pins. Input current limiters, protection against electrostatic and other disturbances, accessible connectors, isolation, and compatibility with defibrillation environments require separate professional design and testing. The assembly described below is therefore limited to a bench setup with a simulated signal, **without connecting a person**.

One ECG lead means one measured potential difference, not necessarily only one physical electrode. Real devices may use an additional contact to stabilize common-mode potential.

3.3. Power supply

It is useful to separate power into three subsystems:

- Battery and its protected management circuitry.
- Low-noise ADC power supply.
- Power for the microcontroller, wireless communication, and status indicators.

The ESP32-S3's current draw can vary substantially when its radio is operating. Accordingly, the analog and digital sections should not depend on a single, poorly filtered supply.

In the portable version, USB-C charging and measurement must have explicitly defined, validated hardware safety modes. During development, it is reasonable to consider disabling the measurement mode whenever a charging or service cable is connected. That interlock alone does not replace IEC 60601-1 requirements.

4. Order of Assembly for the Bench Prototype

The following sequence can produce a functional system that measures and records synthetic ECG signals before a custom medical PCB is developed.

Step 1 — Prepare the evaluation board. Use the ADS1292ECG-FE with its standard MSP430 controller, or an equivalent official development platform. Check the board revision, completeness of the kit, and documentation. Do not connect anything external to the measurement inputs yet.

Step 2 — Check power and the PC connection. Connect the evaluation board using the manufacturer's prescribed method and confirm that the software detects the controller and ADC. At this stage, use only the internal test circuit, with no electrodes or person connected.

Step 3 — Configure the ADC. Select one measurement channel, a rate of 500 samples/s, and an appropriate test gain. Enable the built-in calibration signal and verify that the ADC returns a periodic sequence with the expected shape.

Step 4 — Verify continuous acquisition. Capture a 10-second recording and then a 30-second recording. Check sample count and order, buffer overflow, amplitude of the internal test signal, and successful storage.

Step 5 — Test with a dedicated ECG simulator. Connect a medical ECG simulator through the laboratory interface provided by the manufacturer. The simulator reproduces a known signal shape and rate, allowing subsequent algorithms to be checked.

Step 6 — Build the custom digital platform. After confirming ADC operation, move to a separate ESP32-S3 board. Add buffering, Bluetooth, memory, indication, and autonomous power. Start testing this platform with an internal digital or calibration signal, not a person.

Step 7 — Design the custom PCB. Install the specialized analog front end (AFE), controller, memory, and power circuitry. Base the measurement path on the manufacturer's validated reference design. The patient-protection section must be developed and reviewed separately by a medical-electronics specialist.

At the end of this stage, the result is a functioning hardware/software system able to demonstrate ECG acquisition and processing with test signals, but **not a medical device approved for clinical use**.

5. Configuring ECG Acquisition

For the first working version, I propose the following parameters:

Parameter	Prototype target
Leads	1
Measurement channels	1
Sampling rate	500 Hz
ADC resolution	24 bits
Recording duration	10 or 30 seconds
Raw sample format	Signed 24-bit
Data transmission	Bluetooth LE
Additional connectivity	Wi-Fi
Acquisition mode	Continuous
Time tracking	Sample counter and timestamps
Heart-rate calculation	Based on R–R intervals
Storage	Raw signal plus metadata

These are proposed project parameters that make the technical requirements more specific; they are not values already approved by the original requirements document.

5.1. What 500 Hz means

At 500 Hz, the ADC makes 500 measurements per second. Two consecutive samples are 2 ms apart.

A 30-second recording contains 15,000 samples.

At three bytes per sample, the signal data alone take approximately 45 KB, excluding metadata. If samples are stored in 32-bit containers, they take 60 KB.

These are modest data volumes. The ESP32-S3 can transmit them using Bluetooth LE, and several hundred recordings do not require exceptionally large storage capacity.

5.2. Organizing acquisition

The controller should operate as a finite-state machine.

If an error or an unusable signal is detected, the system should stop presenting unreliable results as valid while retaining diagnostic information about the fault.

At the beginning of a recording, the controller generates a study identifier, resets the sample counter to zero, records the start time, and begins accumulating data.

Storage, heart-rate calculation, and phone communication should run separately from sample acquisition. Generating a PDF must never block ADC reads.

Example firmware logic:

```
Initialization:
    Start controller
    Check power
    Check ADC availability
    Perform internal self-test
    Set sampling rate to 500 Hz
    Prepare ring buffer
    Enter READY

At recording start:
    Create study ID
    Save start time
    Reset sample counter
    Enter RECORDING

When DRDY is received:
    Read digital sample over SPI
    Check ADC status
    Add sample index
    Place sample in buffer

In a separate task:
    Transmit buffered samples
    Watch for buffer overflow
    Update signal-quality state

At recording end:
    Check data completeness
    Record end time
    Save raw signal
    Enter READY
```

This illustrates the control logic, **not** a ready-made ADS1291 driver. The firmware still requires commands and registers from the IC's technical documentation.

6. Signal Processing and Heart-Rate Calculation

The technical requirements call for suppression of baseline wander, power-line interference, and muscle artifacts.

Two versions of each recording should be retained:

Raw signal: the complete ADC output without application-level digital filtering. This should serve as the digital original of the examination.

Processed signal: the version used for display and automated analysis. The user should be able to switch between the two.

6.1. Proposed filters

Filter	Demonstration setting	Purpose
High-pass	0.5 Hz	Remove slow baseline drift
Low-pass	40 Hz	Suppress higher-frequency noise
Notch	50 or 60 Hz, selectable	Suppress mains interference
No application-level filtering	Required	View raw measurements

The 0.5–40 Hz bandwidth is suitable as an initial rhythm-display mode, but must not automatically be treated as a diagnostically complete passband. Excessive filtering can change the shapes of ECG segments and waves.

A separate, broader, validated mode is needed for waveform morphology assessment.

6.2. Example of digital processing

Python with `scipy` can be used for laboratory analysis:

```
import numpy as np
from scipy.signal import butter, sosfiltfilt, iirnotch, filtfilt

FS = 500


def filter_ecg(samples, fs=FS, notch_enabled=True):
    x = np.asarray(samples, dtype=np.float64)

    # Rhythm-display range: 0.5–40 Hz
    sos = butter(
        4, [0.5, 40.0],
        btype="bandpass",
        fs=fs,
        output="sos",
    )
    y = sosfiltfilt(sos, x)

    if notch_enabled:
        b, a = iirnotch(50.0, Q=30, fs=fs)
        y = filtfilt(b, a, y)

    return y
```

This code is intended for processing **completed laboratory recordings from a simulator**. The `filtfilt` operation uses samples both before and after the point being processed, so real-time streaming requires a different, causal implementation that accounts for processing delay.

For a working medical version, filter coefficients, phase characteristics, possible waveform distortion, and processing parameters must be validated with standardized test signals.

6.3. Heart-rate calculation

Heart rate can be calculated from the intervals between detected QRS complexes, more specifically from the locations of R peaks.

If the R–R interval is one second, heart rate is 60 beats/min.

General formula:

HR = 60 / median(RR)

Here RR denotes the intervals between consecutive R peaks, measured in seconds.

Software peak detection generally involves isolating the frequency band containing the main QRS components, computing indicators of rapid signal change, locating candidate peaks, checking their timing, and rejecting artifacts.

A tested algorithm from the Pan–Tompkins family can be used, but it does not become medically validated merely because it works on a few test recordings.

If signal quality is inadequate, the application should display **“Heart rate unavailable”** rather than return an arbitrary number.

6.4. Signal-quality monitoring

Signal-quality monitoring is best implemented as a separate module rather than being mixed with filtering and heart-rate calculation.

It should watch for ADC saturation, breaks in the sample sequence, excessive interference, unstable baseline, and electrode lead-off reports from the measurement front end.

I propose three states:

- **GOOD:** data are suitable for the intended processing.
- **POOR:** quality is reduced; a warning is required.
- **INVALID:** data are unreliable or absent.

The thresholds for these states cannot yet be justified because the technical requirements do not specify numerical limits for noise or interference immunity. They must be established and verified through testing.

7. Bluetooth Data Transmission

For the portable version, Bluetooth Low Energy (BLE) is a logical primary communication channel. Wi-Fi can be reserved for transferring completed examinations through an application or server.

The ESP32-S3 operates as a BLE peripheral. A computer or smartphone connects to it and receives measurement packets.

Each packet should contain the study number or a reference to it, the index of the first sample, a sequence of measurements, and registration-status information.

Simplified packet layout:

```
ECG packet
|-- protocol_version
|-- session_id
|-- first_sample_index
|-- samples_count
|-- signal_samples
|-- quality_flags
`-- checksum
```

The `first_sample_index` field enables the computer to detect missing packets. If it receives a packet beginning at sample 1000 and the next begins at 1100 when 1020 was expected, it must not silently join the two fragments with a line.

Provide for retransmission or explicit marking of lost data. A real patient-facing device will also require secure pairing or connections, authentication, and protection against unauthorized access.

8. Software on the Computer

For the initial prototype, the application can be developed in Python.

Proposed technology stack:

Task	Technology
Application UI	PySide6
Plotting	PyQtGraph
USB data reception	pyserial
BLE communication	bleak
Numerical processing	NumPy, SciPy
CSV export	pandas
PDF generation	ReportLab
Local database	SQLite
Server API	FastAPI
Server database	PostgreSQL

8.1. Proposed application window

An illustrative interface could display:

```
ECG Monitor                      DEMO

Heart rate       72 bpm
Recording        00:10 / 00:30
Sampling rate    500 Hz

Lead I — simulated signal          GOOD
[Illustrative ECG waveform]

Mode: test recording       Filters: 0.5–40 Hz

[Start Recording]            [Save PDF]
```

The waveform is schematic, not a real patient recording. This mock-up illustrates the placement of elements in the future application; its controls are not functional.

8.2. Application logic

The application should contain separate components:

- `DeviceManager` discovers the device, establishes the connection, and tracks link status.
- `ECGReceiver` receives packets, reconstructs the sample sequence, and detects losses.
- `SignalProcessor` performs filtering.
- `HeartRateAnalyzer` calculates heart rate.
- `RecordingManager` saves the examination.
- `ReportGenerator` prepares PDF and the other export formats.

A stable plot need not be redrawn 500 times per second. All 500 samples per second may be acquired while the display is refreshed at approximately 20–30 frames per second.

No data are discarded from the stored recording. Only the acquisition and screen-refresh rates differ.

9. Examination Storage and Telemedicine Integration

9.1. Examination structure

For every examination, save the recording metadata, measurement settings, and signal itself.

Example metadata:

```
{
  "study_id": "ECG-000001",
  "patient_id": "PAT-00001",
  "device_id": "DEV-00001",
  "lead": "I",
  "sample_rate": 500,
  "adc_bits": 24,
  "duration_seconds": 30,
  "samples_count": 15000,
  "signal_unit": "mV",
  "signal_quality": "GOOD",
  "recording_type": "SIMULATED",
  "filter_profile": "rhythm_0.5_40",
  "firmware_version": "0.1.0"
}
```

This structure is for data-format testing only; the identifiers are fictional.

Also save the date/time including time zone, gain and conversion settings for physical units, calibration information, the processing-software version, and an integrity check of the original file.

9.2. Formats

Format	Content
CSV	ECG samples, time, quality, and optionally raw ADC codes
PDF	Waveform, recording settings, examination information
XML	Structured data and metadata
SCP-ECG	Standardized exchange of ECG data
DICOM	Integration with medical archives and information systems

I would divide development into two levels. In the initial prototype, implement CSV, PDF, and a custom, documented XML schema. Once the measurement front end has been validated, implement full interoperability with the data-exchange standards selected together with the target medical information system.

An ordinary XML file containing ECG data does **not** automatically conform to SCP-ECG or DICOM.

9.3. Transmission to a physician

A server architecture may follow this sequence:

```
ECG device
    |
Bluetooth LE
    |
PC / phone application
    |
 HTTPS / API
    |
Study server
    |
Metadata database + signal storage
    |
Physician web interface
    |
HIS / EHR export
```

Each examination should have a unique ID on the server. Patient information should be stored separately from the sample array and linked through identifiers with access controls.

Minimum REST API outline:

Method	Endpoint	Purpose
POST	/api/studies	Create a study
POST	/api/studies/{id}/signal	Upload signal
GET	/api/studies/{id}	Retrieve metadata
GET	/api/studies/{id}/ecg	Retrieve recording
GET	/api/studies/{id}/report.pdf	Retrieve report

The system requires user authorization, role-based access control, audit logging of data access, and protection of medical information both in transit and at rest.

9.4. Use of AI

The technical requirements allow for remote interpretation by a physician or AI.

It makes sense to implement that capability separately from the basic measurement device. First, obtain reproducible digital recordings; then evaluate recognition algorithms against labeled ECG datasets with independent clinical review.

A diagnostic conclusion must not be based solely on an algorithm correctly calculating heart rate. Arrhythmia detection, beat classification, and medical interpretation require their own validation.

10. Testing the Finished Device

The supplied requirements list amplitude accuracy, heart-rate accuracy, noise, filtering, data transfer, and lead-off detection as subjects for verification, but they do not establish numerical pass/fail criteria.

For the bench stage, I propose the following plan:

Test	Method	Preliminary prototype criterion
Sampling	30-second recording	15,000 samples
Data integrity	Check sample numbering	No unreported losses
Amplitude	Calibrated test signal	Error no greater than 5% — provisional goal
Heart rate	Simulate 60, 90, 120 bpm	Error no greater than 2 bpm — provisional goal
Filters	Reference signal at known frequencies	Measured frequency response matches the target
Quality	Introduce simulated interference	Appropriate warning appears
Contact	Simulate a disconnected electrode	Corresponding flag is correctly set
Bluetooth	Transfer a 30-second recording	No undetected losses
CSV	Re-import exported data	All saved samples match
PDF	Review generated report	Correct examination information and parameters
Power	Bench discharge test	Measured battery life meets the agreed target

The figures of 5% and 2 bpm are **suggested engineering debugging targets**, not regulatory tolerances for a medical electrocardiograph. Final acceptance criteria depend on the device's intended purpose and the applicable standard.

Long-duration operation, restart, Bluetooth disconnection, memory overflow, recovery after failure, and correct status messages also require testing.

10.1. Noise measurement

Initially, noise can be assessed using the internally shorted input mode available on specialized ADCs. This helps separate the electronics' intrinsic noise from external interference and contact quality.

Calculate root-mean-square (RMS) noise, peak-to-peak noise, and, if necessary, noise spectral density. State the measurement bandwidth, sampling rate, and gain.

A statement such as “8 µV noise” without bandwidth and measurement method is almost useless for assessing compliance with the requirements.

10.2. Tests that bench checks cannot replace

Defibrillator protection, leakage currents, dielectric strength of isolation, electromagnetic immunity, and safety under component failures require specialized testing facilities.

In particular, the existence of electrostatic-discharge protection in an IC does **not** establish defibrillator protection. Defibrillation pulse immunity must be verified professionally rather than through informal bench testing.

11. PCB and Enclosure

A custom PCB is needed after successful bench prototyping.

I would design the board in KiCad or Altium Designer. A four-layer design is worth considering for this device, to support controlled return-current paths, stable power, and reduced interference.

Component placement should account for the distance between sensitive analog traces and the ESP32 radio, decoupling capacitor placement, correct ground organization, separation of noisy digital circuits from the low-noise analog section, and related layout details.

Avoid routing high-speed digital traces close to ADC inputs or using long, unprotected analog connections. Begin with the manufacturer's recommended schematic and PCB layout, then evaluate any changes.

The enclosure should provide secure PCB mounting, access to controls, visible indicators, a protected USB-C port, labeling, mechanical robustness, and battery maintenance provisions. The medical version also requires assessment of safety clearances, materials, and user-accessible parts.

An initial battery-life goal of at least eight hours can be proposed, but cell selection and capacity must follow real measurements of ESP32 current consumption in the intended Bluetooth/Wi-Fi modes. The original requirements do not define a particular number of hours.

12. Applicable Standards

A key feature of the technical requirements is that the device's intended purpose must be specified more precisely before the complete set of applicable standards can be determined.

At present, it is described as a device for 10–30-second ECG recordings for screening and communication with a physician. That does not necessarily make it a continuous ambulatory monitor.

Standard	Scope
IEC 60601-1	General safety of medical electrical equipment
IEC 60601-1-2	Electromagnetic compatibility
IEC 60601-1-11	Additional requirements for the home healthcare environment
IEC 60601-2-25	Electrocardiographs intended to produce diagnostic records
IEC 60601-2-27	ECG monitoring equipment, within its defined scope
IEC 60601-2-47	Continuous ambulatory ECG acquisition and analysis systems
IEC 62304	Medical-device software life-cycle processes
ISO 14971:2019	Medical-device risk management
ISO 10993-1:2025	Biological evaluation of patient-contacting materials

IEC 60601-2-47 expressly excludes systems that do not provide continuous recording and analysis, such as event recorders. Therefore, applying it to a 10–30-second ECG device merely because that device is “portable” would be inappropriate. The relevant particular standard depends on the declared intended purpose.

For implementation in Kazakhstan, the device's risk class and regulatory registration pathway must also be established. According to published information from Kazakhstan's government authorities, the transitional option of registering medical devices under national rules alongside EAEU rules has been extended through **31 December 2027**.

Technical documentation, risk analysis, software documentation, verification and validation records, operating documents, and safety evidence must be prepared in advance.

13. Final Development Sequence

Stage A — Software prototype. Develop a synthetic ECG generator and implement plotting, filters, heart-rate calculation, and CSV/PDF saving. Check the algorithms against known test sequences.

Stage B — Measurement bench. Use the ADS1292ECG-FE in single-channel mode. Check digitization of its built-in test signal and a dedicated ECG simulator.

Stage C — Portable electronics. Implement the controller, communication protocol, autonomous power, memory, and status indicators. Verify operation on the bench.

Stage D — Telemedicine. Add a local archive, server, access management, physician data transfer, and the necessary interchange formats for medical systems.

Stage E — Medical device. Complete the measurement front end, patient protection, construction, software, and documentation. Perform independent safety tests and the applicable regulatory procedures.

In summary: for this task, I would use the ADS1291 as the basis for the final single-channel device, the ADS1292ECG-FE for laboratory acquisition development, the ESP32-S3 for communication and control, and Python/PySide6 for the application and initial telemedicine infrastructure.

This architecture makes it possible to check almost all of the functional requirements before investing in an expensive custom PCB. However, connecting the device to a person must remain a separate stage, possible only after electrical safety has been confirmed: accurate acquisition of a simulated signal does not establish patient safety.